

RANDRIAMIALISOA Antsa Ainatiavina Joycia

183125 L2 INFO

MYSQL

Projet BD_AIR_MAROC :

-D'abord on crée base de donnée BD_AIR_MAROC:

```
MariaDB [(none)]> CREATE DATABASE BD_AIR_MAROC;  
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [(none)]> SHOW DATABASES;
```

Database
bd_air_maroc

-Passons au création des tables et l'affichage des description :

CREATION PILOTE :

```
MariaDB [bd_air_maroc]> CREATE TABLE PILOTE (  
-> NUMPIL INT AUTO_INCREMENT PRIMARY KEY,  
-> NOMPIL VARCHAR(100) NOT NULL,  
-> Ville VARCHAR(50) NOT NULL,  
-> Salaire DECIMAL(10, 2) NOT NULL  
-> ) AUTO_INCREMENT = 1;  
Query OK, 0 rows affected (0.011 sec)
```

DESCRIPTION :

```
MariaDB [bd_air_maroc]> desc PILOTE;
```

Field	Type	Null	Key	Default	Extra
NUMPIL	int(11)	NO	PRI	NULL	auto_increment
NOMPIL	varchar(100)	NO		NULL	
Ville	varchar(50)	NO		NULL	
Salaire	decimal(10,2)	NO		NULL	

4 rows in set (0.026 sec)

CREATION VOL

```
MariaDB [bd_air_maroc]> CREATE TABLE Avion (  
-> NUMAV INT AUTO_INCREMENT PRIMARY KEY,  
-> Nomav VARCHAR(50) NOT NULL,  
-> Capacite INT NOT NULL,  
-> Ville VARCHAR(50) NOT NULL  
-> ) AUTO_INCREMENT = 80;  
Query OK, 0 rows affected (0.011 sec)
```

```
MariaDB [bd_air_maroc]> desc VOL;
```

Field	Type	Null	Key	Default	Extra
Numvol	varchar(10)	NO	PRI	NULL	
NUMPIL	int(11)	NO	MUL	NULL	
NUMAV	int(11)	NO	MUL	NULL	
Ville_dep	varchar(50)	NO		NULL	
Ville_arr	varchar(50)	NO		NULL	
H_dep	time	NO		NULL	
H_arr	time	YES		NULL	

7 rows in set (0.019 sec)

CREATION AVION

```
MariaDB [bd_air_maroc]> Create table AVION(  
-> NUMAV INT AUTO_INCREMENT PRIMARY KEY ,  
-> Nomav VARCHAR(50),  
-> Capacite INT NOT NULL,  
-> Ville VARCHAR(50) NOT NULL  
-> ) AUTO_INCREMENT = 10;  
Query OK, 0 rows affected (0.013 sec)
```

```
MariaDB [bd_air_maroc]> desc AVION;
```

Field	Type	Null	Key	Default	Extra
NUMAV	int(11)	NO	PRI	NULL	auto_increment
Nomav	varchar(50)	YES		NULL	
Capacite	int(11)	NO		NULL	
Ville	varchar(50)	NO		NULL	

4 rows in set (0.020 sec)

-INSERTION DES DONNEES

AVION

```
MariaDB [bd_air_maroc]> INSERT INTO Avion(Nomav,Capacite,Ville) VALUES('Airbus A320', 180, 'Casablanca'), ('Boeing 737', 150, 'Marrakech'), ('ATR 72', 72,  
-> 'Agadir'),  
-> ('Airbus A350', 300, 'Tanger'), ('Falcon 900', 12, 'Rabat'), ('Boeing 747', 400, 'Fes'),  
-> ('Airbus A321', 200, 'Oujda'), ('Boeing 777', 350, 'Meknes'), ('ATR 42', 48, 'Guelmin'),  
-> ('Airbus A320', 180, 'Casablanca'), ('Boeing 737', 150, 'Marrakech'), ('ATR 72', 72,  
-> 'Agadir'),  
-> ('Airbus A350', 300, 'Tanger'), ('Falcon 900', 12, 'Rabat'), ('Boeing 747', 400, 'Fes'),  
-> ('Airbus A321', 200, 'Oujda'), ('Boeing 777', 350, 'Meknes'), ('ATR 42', 48, 'Guelmin'),  
-> ('Airbus A320', 180, 'Casablanca'), ('Boeing 737', 150, 'Marrakech'), ('ATR 72', 72,  
-> 'Agadir'),  
-> ('Airbus A350', 300, 'Tanger'), ('Falcon 900', 12, 'Rabat'), ('Boeing 747', 400, 'Fes'),  
-> ('Airbus A321', 200, 'Oujda'), ('Boeing 777', 350, 'Meknes'), ('ATR 42', 48, 'Guelmin'),  
-> ('Airbus A320', 180, 'Casablanca'), ('Boeing 737', 150, 'Marrakech'), ('ATR 72', 72,  
-> 'Agadir'),  
-> ('Airbus A350', 300, 'Tanger'), ('Falcon 900', 12, 'Rabat'), ('Boeing 747', 400, 'Fes'),  
-> ('Airbus A321', 200, 'Oujda'), ('Boeing 777', 350, 'Meknes'), ('ATR 42', 48, 'Guelmin'),  
-> ('Airbus A320', 180, 'Casablanca'), ('Boeing 737', 150, 'Marrakech'), ('ATR 72', 72,  
-> 'Agadir'),  
-> ('Airbus A350', 300, 'Tanger'), ('Falcon 900', 12, 'Rabat'), ('Boeing 747', 400, 'Fes');  
Query OK, 51 rows affected (0.004 sec)  
Records: 51 Duplicates: 0 Warnings: 0
```

PILOTE

```
MariaDB [bd_air_maroc]> INSERT INTO PILOTE(NOMPIL,Ville,Salaire) VALUES('Sami', 'Marrakech', 11450), ('Amine', 'Meknes', 31200), ('Laila', 'Guelmin', 28500),
-> ('Yassine', 'Casablanca', 12000), ('Houda', 'Rabat', 14100), ('Omar', 'Tanger', 19300),
-> ('Zineb', 'Agadir', 5600), ('Karim', 'Fes', 10800), ('Anas', 'Oujda', 13000),
-> ('Sara', 'Marrakech', 7400), ('Mahdi', 'Meknes', 11900), ('Noura', 'Guelmin', 16200),
-> ('Tarik', 'Casablanca', 3500), ('Salma', 'Rabat', 18800), ('Hamza', 'Tanger', 14700),
-> ('Ines', 'Agadir', 10200), ('Walid', 'Fes', 15900), ('Ghita', 'Oujda', 9600),
-> ('Rayan', 'Marrakech', 30150), ('Sofia', 'Meknes', 11100), ('Adil', 'Guelmin', 14300),
-> ('Kenza', 'Casablanca', 7900), ('Faycal', 'Rabat', 15200), ('Malak', 'Tanger', 11500),
-> ('Saïd', 'Agadir', 13800), ('Asma', 'Fes', 9100), ('Jalil', 'Oujda', 6700),
-> ('Imane', 'Marrakech', 10500), ('Youssef', 'Meknes', 14900), ('Meryem', 'Guelmin', 8200),
-> ('Khalid', 'Casablanca', 3300), ('Lamia', 'Rabat', 11800), ('Brahim', 'Tanger', 15400),
-> ('Hajar', 'Agadir', 19700), ('Ismail', 'Fes', 14000), ('Rania', 'Oujda', 18600),
-> ('Taha', 'Marrakech', 11200), ('Dounia', 'Meknes', 3700), ('Nabil', 'Guelmin', 17100),
-> ('Aïcha', 'Casablanca', 16000), ('Zakaria', 'Rabat', 10900), ('Lina', 'Tanger', 14500),
-> ('Adam', 'Agadir', 18300), ('Sana', 'Fes', 34000), ('Ilyas', 'Oujda', 11700),
-> ('Nada', 'Marrakech', 5100), ('Reda', 'Meknes', 19400), ('Rim', 'Guelmin', 14800),
-> ('Fouad', 'Casablanca', 10600), ('Soukaina', 'Rabat', 30900);
Query OK, 50 rows affected (0.004 sec)
```

VOL

```
MariaDB [bd_air_maroc]> INSERT INTO Vol(Numvol,NUMPIL,NUMAV,Ville_dep,Ville_arr,H_dep,H_arr) VALUES('V102', 2, 81, 'Meknes', 'Rabat', '09:30', '10:15'),
-> ('V103', 3, 82, 'Guelmin', 'Agadir', '11:00', '12:00'),
-> ('V104', 4, 83, 'Casablanca', 'Tanger', '14:00', '15:10'),
-> ('V105', 3, 84, 'Rabat', 'Fes', '16:00', '16:45'),
-> ('V106', 6, 81, 'Tanger', 'Oujda', '18:00', '19:30'),
-> ('V107', 7, 82, 'Agadir', 'Marrakech', '07:15', '08:00'),
-> ('V108', 8, 83, 'Fes', 'Meknes', '10:00', '10:30'),
-> ('V109', 9, 84, 'Oujda', 'Casablanca', '12:30', '14:00'),
-> ('V110', 11, 89, 'Marrakech', 'Agadir', '15:00', '15:45'),
-> ('V111', 11, 96, 'Meknes', 'Tanger', '06:00', '07:30'),
-> ('V112', 2, 91, 'Guelmin', 'Casablanca', '08:30', '10:00'),
-> ('V113', 3, 92, 'Casablanca', 'Rabat', '11:00', '11:40'),
-> ('V114', 3, 90, 'Rabat', 'Marrakech', '13:00', '14:15'),
-> ('V115', 9, 94, 'Tanger', 'Agadir', '17:00', '18:45'),
-> ('V116', 16, 95, 'Agadir', 'Fes', '19:00', '20:30'),
-> ('V117', 7, 92, 'Fes', 'Oujda', '21:00', '22:00'),
-> ('V118', 18, 97, 'Oujda', 'Marrakech', '05:30', '07:15'),
-> ('V119', 19, 98, 'Marrakech', 'Meknes', '09:00', '10:00'),
-> ('V120', 10, 99, 'Meknes', 'Guelmin', '12:00', '13:30'),
-> ('V121', 21, 100, 'Guelmin', 'Rabat', '14:30', '15:50'),
-> ('V122', 22, 101, 'Casablanca', 'Agadir', '16:00', '17:15'),
-> ('V123', 11, 102, 'Rabat', 'Tanger', '18:30', '19:30'),
-> ('V124', 24, 102, 'Tanger', 'Casablanca', '20:00', '21:15'),
-> ('V125', 25, 104, 'Agadir', 'Oujda', '22:30', '23:55'),
-> ('V126', 26, 105, 'Fes', 'Marrakech', '06:45', '07:50'),
-> ('V127', 17, 104, 'Oujda', 'Agadir', '08:15', '09:45'),
-> ('V128', 28, 107, 'Marrakech', 'Rabat', '10:00', '11:15'),
-> ('V129', 29, 106, 'Meknes', 'Fes', '12:00', '12:30'),
-> ('V130', 10, 109, 'Guelmin', 'Tanger', '14:00', '16:00'),
-> ('V131', 31, 110, 'Casablanca', 'Marrakech', '15:30', '16:30'),
-> ('V132', 32, 101, 'Rabat', 'Agadir', '17:15', '18:30'),
-> ('V133', 13, 110, 'Tanger', 'Meknes', '19:00', '20:45'),
-> ('V134', 34, 103, 'Agadir', 'Casablanca', '21:00', '22:15'),
-> ('V135', 35, 114, 'Fes', 'Guelmin', '23:00', '00:30'),
-> ('V136', 26, 115, 'Oujda', 'Rabat', '07:00', '08:15'),
-> ('V137', 37, 126, 'Marrakech', 'Fes', '09:15', '10:45'),
-> ('V138', 38, 117, 'Meknes', 'Casablanca', '11:30', '12:45'),
-> ('V139', 39, 128, 'Guelmin', 'Marrakech', '13:45', '14:45'),
-> ('V140', 4, 119, 'Casablanca', 'Oujda', '15:00', '16:30'),
-> ('V141', 41, 120, 'Rabat', 'Agadir', '17:30', '18:45'),
-> ('V142', 42, 121, 'Tanger', 'Fes', '19:45', '20:45'),
-> ('V143', 3, 122, 'Agadir', 'Rabat', '06:00', '07:15'),
-> ('V144', 44, 123, 'Fes', 'Casablanca', '08:30', '09:45'),
-> ('V145', 45, 124, 'Oujda', 'Tanger', '10:45', '12:15'),
-> ('V146', 46, 125, 'Marrakech', 'Guelmin', '13:00', '14:20'),
-> ('V147', 7, 126, 'Meknes', 'Agadir', '15:15', '16:45'),
-> ('V148', 48, 127, 'Guelmin', 'Fes', '17:30', '19:00'),
-> ('V149', 49, 128, 'Casablanca', 'Rabat', '20:00', '20:45'),
-> ('V150', 50, 129, 'Rabat', 'Oujda', '22:00', '23:30');
Query OK, 49 rows affected (0.009 sec)
Records: 49 Duplicates: 0 Warnings: 0
```

REPONSES AUX REQUETE

1. Liste des avions dont la capacité est supérieure à 350 passagers :

```
MariaDB [bd_air_maroc]> SELECT * FROM AVION WHERE CAPACITE > 350;
+-----+-----+-----+-----+
| NUMAV | Nomav | Capacte | Ville |
+-----+-----+-----+-----+
| 85 | Boeing 747 | 400 | Fes |
| 94 | Boeing 747 | 400 | Fes |
| 103 | Boeing 747 | 400 | Fes |
| 112 | Boeing 747 | 400 | Fes |
| 121 | Boeing 747 | 400 | Fes |
| 130 | Boeing 747 | 400 | Fes |
+-----+-----+-----+-----+
6 rows in set (0.001 sec)
```

2. Numéros et noms des avions localisés à Marrakech :

```
MariaDB [bd_air_maroc]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech';
```

NUMAV	NOMAV
81	Boeing 737
90	Boeing 737
99	Boeing 737
108	Boeing 737
117	Boeing 737
126	Boeing 737

6 rows in set (0.000 sec)

3. Pilotes en service et villes de départ :

```
MariaDB [bd_air_maroc]> SELECT NUMPIL, VILLE_DEP FROM VOL order by NUMPIL ASC;
```

NUMPIL	VILLE_DEP
2	Meknes
2	Guelmin
3	Agadir
3	Rabat
3	Rabat
3	Guelmin
3	Casablanca
4	Casablanca
4	Casablanca
6	Tanger
7	Meknes
7	Fes
7	Agadir
8	Fes
9	Tanger
9	Oujda
10	Guelmin
10	Meknes
11	Marrakech
11	Rabat
11	Meknes
13	Tanger
16	Agadir
17	Oujda
18	Oujda
19	Marrakech
21	Guelmin
22	Casablanca
24	Tanger
25	Agadir
26	Oujda
26	Fes
28	Marrakech
29	Meknes
31	Casablanca
32	Rabat
34	Agadir

35	Fes
37	Marrakech
38	Meknes
39	Guelmin
41	Rabat
42	Tanger
44	Fes
45	Oujda
46	Marrakech
48	Guelmin
49	Casablanca
50	Rabat

49 rows in set (0.000 sec)

4. Toutes les informations sur les pilotes :


```
MariaDB [bd_air_maroc]> SELECT * FROM PILOTE;
```

NUMPIL	NOMPIL	Ville	Salaire
1	Sami	Marrakech	11450.00
2	Amine	Meknes	31200.00
3	Laila	Guelmin	28500.00
4	Yassine	Casablanca	12000.00
5	Houda	Rabat	14100.00
6	Omar	Tanger	19300.00
7	Zineb	Agadir	5600.00
8	Karim	Fes	10800.00
9	Anas	Oujda	13000.00
10	Sara	Marrakech	7400.00
11	Mehdi	Meknes	11900.00
12	Noura	Guelmin	16200.00
13	Tarik	Casablanca	3500.00
14	Salma	Rabat	18800.00
15	Hamza	Tanger	14700.00
16	Ines	Agadir	10200.00
17	Walid	Fes	15900.00
18	Ghita	Oujda	9600.00
19	Rayan	Marrakech	30150.00
20	Sofia	Meknes	11100.00
21	Adil	Guelmin	14300.00
22	Kenza	Casablanca	7900.00
23	Faycal	Rabat	15200.00
24	Malak	Tanger	11500.00
25	Said	Agadir	13800.00
26	Asma	Fes	9100.00
27	Jalil	Oujda	6700.00
28	Imane	Marrakech	10500.00
29	Youssef	Meknes	14900.00
30	Meryem	Guelmin	8200.00
31	Khalid	Casablanca	3300.00
32	Lamia	Rabat	11800.00
33	Brahim	Tanger	15400.00
34	Hajar	Agadir	19700.00
35	Ismail	Fes	14000.00
36	Rania	Oujda	18600.00
37	Taha	Marrakech	11200.00
38	Dounia	Meknes	3700.00
39	Nabil	Guelmin	17100.00
40	Aicha	Casablanca	16000.00
41	Zakaria	Rabat	10900.00
42	Lina	Tanger	14500.00
43	Adam	Agadir	18300.00
44	Sana	Fes	34000.00
45	Ilyas	Oujda	11700.00
46	Nada	Marrakech	5100.00
47	Reda	Meknes	19400.00
48	Rim	Guelmin	14800.00
49	Fouad	Casablanca	10600.00
50	Soukaina	Rabat	30900.00

```
50 rows in set (0.000 sec)
```

5. Pilotes domiciliés à Meknès avec salaire > 20 000 DH :

```
MariaDB [bd_air_maroc]> SELECT NOMPIL FROM PILOTE WHERE VILLE = 'Meknes' AND SALAIRE > 20000;
```

```
+-----+  
| NOMPIL |  
+-----+  
| Amine  |  
+-----+
```

```
1 row in set (0.000 sec)
```

6. Avions à Marrakech ou capacité < 350 :

```
MariaDB [bd_air_maroc]> SELECT NUMAV, NOMAV FROM AVION WHERE VILLE = 'Marrakech' OR CAPACITE < 350;
+-----+
| NUMAV | NOMAV |
+-----+
| 80    | Airbus A320 |
| 81    | Boeing 737  |
| 82    | ATR 72      |
| 83    | Airbus A350 |
| 84    | Falcon 900  |
| 86    | Airbus A321 |
| 88    | ATR 42      |
| 89    | Airbus A320 |
| 90    | Boeing 737  |
| 91    | ATR 72      |
| 92    | Airbus A350 |
| 93    | Falcon 900  |
| 95    | Airbus A321 |
| 97    | ATR 42      |
| 98    | Airbus A320 |
| 99    | Boeing 737  |
| 100   | ATR 72      |
| 101   | Airbus A350 |
| 102   | Falcon 900  |
| 104   | Airbus A321 |
| 106   | ATR 42      |
| 107   | Airbus A320 |
| 108   | Boeing 737  |
| 109   | ATR 72      |
| 110   | Airbus A350 |
| 111   | Falcon 900  |
| 113   | Airbus A321 |
| 115   | ATR 42      |
| 116   | Airbus A320 |
| 117   | Boeing 737  |
| 118   | ATR 72      |
| 119   | Airbus A350 |
| 120   | Falcon 900  |
| 122   | Airbus A321 |
| 124   | ATR 42      |
| 125   | Airbus A320 |
| 126   | Boeing 737  |
| 127   | ATR 72      |
| 128   | Airbus A350 |
| 129   | Falcon 900  |
+-----+
40 rows in set (0.000 sec)
```

7. Pilotes pas en service :

```
MariaDB [bd_air_maroc]> SELECT NUMPIL FROM PILOTE WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
+-----+
| NUMPIL |
+-----+
| 1      |
| 5      |
| 12     |
| 14     |
| 15     |
| 20     |
| 23     |
| 27     |
| 30     |
| 33     |
| 36     |
| 40     |
| 43     |
| 47     |
+-----+
14 rows in set (0.001 sec)
```

8. Vols au départ de Marrakech par pilotes de Meknès :

```
MariaDB [bd_air_maroc]> SELECT NUMVOL FROM VOL V JOIN PILOTE P ON V.NUMPIL = P.NUMPIL WHERE V.VILLE_DEP =
-> 'Marrakech' AND P.VILLE = 'Meknes';
+-----+
| NUMVOL |
+-----+
| V110   |
+-----+
1 row in set (0.000 sec)
```

9. Vols par avion non localisé à Marrakech :

```
MariaDB [bd_air_maroc]> SELECT NUMVOL FROM VOL V JOIN AVION A ON V.NUMAV = A.NUMAV WHERE A.VILLE <> 'Marrakech';
```

NUMVOL
V103
V107
V104
V108
V105
V109
V110
V112
V113
V117
V115
V116
V111
V118
V119
V121
V122
V132
V123
V124
V134
V125
V127
V126
V129
V128
V130
V131
V133
V135
V136
V140
V141
V142
V143
V144
V145
V146
V148
V139
V149
V150

```
42 rows in set (0.000 sec)
```

10. Villes après Guelmim :

```
MariaDB [bd_air_maroc]> SELECT DISTINCT VILLE_ARR FROM VOL WHERE VILLE_DEP IN (SELECT VILLE_ARR FROM VOL WHERE  
-> VILLE_DEP = 'Guelmim');  
Empty set (0.001 sec)
```

11. Pilotes et numéros de vols :

```
MariaDB [bd_air_maroc]> SELECT NOMPIL, NUMVOL FROM PILOTE P JOIN VOL V ON P.NUMPIL = V.NUMPIL;
```

NOMPIL	NUMVOL
Amine	V102
Amine	V112
Laila	V103
Laila	V105
Laila	V113
Laila	V114
Laila	V143
Yassine	V104
Yassine	V140
Omar	V106
Zineb	V107
Zineb	V117
Zineb	V147
Karim	V108
Anas	V109
Anas	V115
Sara	V120
Sara	V130
Mehdi	V110
Mehdi	V111
Mehdi	V123
Tarik	V133
Ines	V116
Walid	V127
Ghita	V118
Rayan	V119
Adil	V121
Kenza	V122
Malak	V124
Said	V125
Asma	V126
Asma	V136
Imane	V128
Youssef	V129
Khalid	V131
Lamia	V132
Hajar	V134
Ismail	V135
Taha	V137
Dounia	V138
Nabil	V139
Zakaria	V141
Lina	V142
Sana	V144
Ilyas	V145
Nada	V146
Rim	V148
Fouad	V149
Soukaina	V150

49 rows in set (0.001 sec)

12. Noms pilotes et noms avions pilotés :

```
MariaDB [bd_air_maroc]> SELECT DISTINCT NOMPIL, NOMAV FROM PILOTE P JOIN VOL V ON P.NUMPIL = V.NUMPIL JOIN AVION A ON V.NUMAV = A.NUMAV;
```

NOMPIL	NOMAV
Amine	Boeing 737
Laila	ATR 72
Yassine	Airbus A350
Laila	Falcon 900
Omar	Boeing 737
Zineb	ATR 72
Karim	Airbus A350
Anas	Falcon 900
Mehdi	Airbus A320
Mehdi	Boeing 777
Amine	ATR 72
Laila	Airbus A350
Laila	Boeing 737
Anas	Boeing 747
Ines	Airbus A321
Zineb	Airbus A350
Ghita	ATR 42
Rayan	Airbus A320
Sara	Boeing 737
Adil	ATR 72
Kenza	Airbus A350
Mehdi	Falcon 900
Malak	Falcon 900
Said	Airbus A321
Asma	Boeing 777
Walid	Airbus A321
Imane	Airbus A320
Youssef	ATR 42
Sara	ATR 72
Khalid	Airbus A350
Lamia	Airbus A350
Tarik	Airbus A350
Hajar	Boeing 747
Ismail	Boeing 777
Asma	ATR 42
Taha	Boeing 737
Dounia	Boeing 737
Nabil	Airbus A350
Zakaria	Falcon 900
Lina	Boeing 747
Laila	Airbus A321
Sana	Boeing 777
Ilyas	ATR 42
Nada	Airbus A320
Zineb	Boeing 737
Rim	ATR 72
Fouad	Airbus A350
Soukaina	Falcon 900

48 rows in set (0.001 sec)

13. Vols avec nom du pilote :

```
MariaDB [bd_air_maroc]> SELECT NUMVOL, VILLE_DEP, VILLE_ARR, NOMPIL FROM VOL V JOIN PILOTE P ON V.NUMPIL =
-> P.NUMPIL;
```

NUMVOL	VILLE_DEP	VILLE_ARR	NOMPIL
V102	Meknes	Rabat	Amine
V103	Guelmin	Agadir	Laila
V104	Casablanca	Tanger	Yassine
V105	Rabat	Fes	Laila
V106	Tanger	Oujda	Omar
V107	Agadir	Marrakech	Zineb
V108	Fes	Meknes	Karim
V109	Oujda	Casablanca	Anas
V110	Marrakech	Agadir	Mehdi
V111	Meknes	Tanger	Mehdi
V112	Guelmin	Casablanca	Amine
V113	Casablanca	Rabat	Laila
V114	Rabat	Marrakech	Laila
V115	Tanger	Agadir	Anas
V116	Agadir	Fes	Ines
V117	Fes	Oujda	Zineb
V118	Oujda	Marrakech	Ghita
V119	Marrakech	Meknes	Rayan
V120	Meknes	Guelmin	Sara
V121	Guelmin	Rabat	Adil
V122	Casablanca	Agadir	Kenza
V123	Rabat	Tanger	Mehdi
V124	Tanger	Casablanca	Malak
V125	Agadir	Oujda	Said
V126	Fes	Marrakech	Asma
V127	Oujda	Agadir	Walid
V128	Marrakech	Rabat	Imane
V129	Meknes	Fes	Youssef
V130	Guelmin	Tanger	Sara
V131	Casablanca	Marrakech	Khalid
V132	Rabat	Agadir	Lamia
V133	Tanger	Meknes	Tarik
V134	Agadir	Casablanca	Hajar
V135	Fes	Guelmin	Ismail
V136	Oujda	Rabat	Asma
V137	Marrakech	Fes	Taha
V138	Meknes	Casablanca	Dounia
V139	Guelmin	Marrakech	Nabil
V140	Casablanca	Oujda	Yassine
V141	Rabat	Agadir	Zakaria
V142	Tanger	Fes	Lina
V143	Agadir	Rabat	Laila
V144	Fes	Casablanca	Sana
V145	Oujda	Tanger	Ilyas
V146	Marrakech	Guelmin	Nada
V147	Meknes	Agadir	Zineb
V148	Guelmin	Fes	Rim
V149	Casablanca	Rabat	Fouad
V150	Rabat	Oujda	Soukaina

```
49 rows in set (0.000 sec)
```

14. Avions au départ de Casablanca :

```
MariaDB [bd_air_maroc]> SELECT DISTINCT NOMAV FROM AVION A JOIN VOL V ON A.NUMAV = V.NUMAV WHERE V.VILLE_DEP =
-> 'Casablanca';
```

NOMAV
Airbus A350

```
1 row in set (0.001 sec)
```


15. Pilotes d'avions de capacité > 300 :

```
MariaDB [bd_air_maroc]> SELECT DISTINCT NOMPIL FROM PILOTE P JOIN VOL V ON P.NUMPIL = V.NUMPIL JOIN AVION A ON  
-> V.NUMAV = A.NUMAV WHERE A.CAPACITE > 300;  
+-----+  
| NOMPIL |  
+-----+  
| Mehdi  |  
| Anas   |  
| Asma   |  
| Hajar  |  
| Ismail |  
| Lina   |  
| Sana   |  
+-----+  
7 rows in set (0.001 sec)
```

16. Pilotes et villes d'arrivée :

```
MariaDB [bd_air_maroc]> SELECT NOMPIL, VILLE_ARR FROM PILOTE P JOIN VOL V ON P.NUMPIL = V.NUMPIL;  
+-----+-----+  
| NOMPIL | VILLE_ARR |  
+-----+-----+  
| Amine  | Rabat     |  
| Laila  | Agadir    |  
| Yassine | Tanger    |  
| Laila  | Fes       |  
| Omar   | Oujda     |  
| Zineb  | Marrakech |  
| Karim  | Meknes    |  
| Anas   | Casablanca |  
| Mehdi  | Agadir    |  
| Mehdi  | Tanger    |  
| Amine  | Casablanca |  
| Laila  | Rabat     |  
| Laila  | Marrakech |  
| Anas   | Agadir    |  
| Ines   | Fes       |  
| Zineb  | Oujda     |  
| Ghita  | Marrakech |  
| Rayan  | Meknes    |  
| Sara   | Guelmin   |  
| Adil   | Rabat     |  
| Kenza  | Agadir    |  
| Mehdi  | Tanger    |  
| Malak  | Casablanca |  
| Said   | Oujda     |  
| Asma   | Marrakech |  
| Walid  | Agadir    |  
| Imane  | Rabat     |  
| Youssef | Fes       |  
| Sara   | Tanger    |  
| Khalid | Marrakech |  
| Lamia  | Agadir    |  
| Tarik  | Meknes    |  
| Hajar  | Casablanca |  
| Ismail | Guelmin   |  
| Asma   | Rabat     |  
| Taha   | Fes       |  
| Dounia | Casablanca |  
| Nabil  | Marrakech |  
| Yassine | Oujda     |  
| Zakaria | Agadir    |  
| Lina   | Fes       |  
| Laila  | Rabat     |  
| Sana   | Casablanca |  
| Ilyas  | Tanger    |  
| Nada   | Guelmin   |  
| Zineb  | Agadir    |  
| Rim    | Fes       |  
| Fouad  | Rabat     |  
| Soukaina | Oujda    |  
+-----+-----+  
49 rows in set (0.001 sec)
```

17. Jointure complète des 3 tables :

```

MariaDB [bd_air_maroc]> SELECT * FROM VOL V JOIN PILOTE P ON V.NUMPIL = P.NUMPIL JOIN AVION A ON V.NUMAV =
-> A.NUMAV;

```

Numvol	NUMPIL	NUMAV	Ville_dep	Ville_arr	H_dep	H_arr	NUMPIL	NOMPIL	Ville	Salaire	NUMAV	Nomav	Capacite	Ville
V102	2	81	Meknes	Rabat	09:30:00	10:15:00	2	Amine	Meknes	31200.00	81	Boeing 737	150	Marrakech
V103	3	82	Guelmin	Agadir	11:00:00	12:00:00	3	Laila	Guelmin	28500.00	82	ATR 72	72	Agadir
V104	4	83	Casablanca	Tanger	14:00:00	15:10:00	4	Yassine	Casablanca	12000.00	83	Airbus A350	300	Tanger
V105	3	84	Rabat	Fes	16:00:00	16:45:00	3	Laila	Guelmin	28500.00	84	Falcon 900	12	Rabat
V106	6	81	Tanger	Oujda	18:00:00	19:30:00	6	Omar	Tanger	19300.00	81	Boeing 737	150	Marrakech
V107	7	82	Agadir	Marrakech	07:15:00	08:00:00	7	Zineb	Agadir	5600.00	82	ATR 72	72	Agadir
V108	8	83	Fes	Meknes	10:00:00	10:30:00	8	Karim	Fes	10800.00	83	Airbus A350	300	Tanger
V109	9	84	Oujda	Casablanca	12:30:00	14:00:00	9	Anas	Oujda	13000.00	84	Falcon 900	12	Rabat
V110	11	89	Marrakech	Agadir	15:00:00	15:45:00	11	Mehdi	Meknes	11900.00	89	Airbus A320	180	Casablanca
V111	11	96	Meknes	Tanger	06:00:00	07:30:00	11	Mehdi	Meknes	11900.00	96	Boeing 777	350	Meknes
V112	2	91	Guelmin	Casablanca	08:30:00	10:00:00	2	Amine	Meknes	31200.00	91	ATR 72	72	Agadir
V113	3	92	Casablanca	Rabat	11:00:00	11:40:00	3	Laila	Guelmin	28500.00	92	Airbus A350	300	Tanger
V114	3	90	Rabat	Marrakech	13:00:00	14:15:00	3	Laila	Guelmin	28500.00	90	Boeing 737	150	Marrakech
V115	9	94	Tanger	Agadir	17:00:00	18:45:00	9	Anas	Oujda	13000.00	94	Boeing 747	400	Fes
V116	16	95	Agadir	Fes	19:00:00	20:30:00	16	Ines	Agadir	10200.00	95	Airbus A321	200	Oujda
V117	7	92	Fes	Oujda	21:00:00	22:00:00	7	Zineb	Agadir	5600.00	92	Airbus A350	300	Tanger
V118	18	97	Oujda	Marrakech	05:30:00	07:15:00	18	Ghita	Oujda	9600.00	97	ATR 42	48	Guelmin
V119	19	98	Marrakech	Meknes	09:00:00	10:00:00	19	Rayan	Marrakech	30150.00	98	Airbus A320	180	Casablanca
V120	10	99	Meknes	Guelmin	12:00:00	13:30:00	10	Sara	Marrakech	7400.00	99	Boeing 737	150	Marrakech
V121	21	100	Guelmin	Rabat	14:30:00	15:50:00	21	Adil	Guelmin	14300.00	100	ATR 72	72	Agadir
V122	22	101	Casablanca	Agadir	16:00:00	17:15:00	22	Kenza	Casablanca	7900.00	101	Airbus A350	300	Tanger
V123	11	102	Rabat	Tanger	18:30:00	19:30:00	11	Mehdi	Meknes	11900.00	102	Falcon 900	12	Rabat
V124	24	102	Tanger	Casablanca	20:00:00	21:15:00	24	Malak	Tanger	11500.00	102	Falcon 900	12	Rabat
V125	25	104	Agadir	Oujda	22:30:00	23:55:00	25	Said	Agadir	13800.00	104	Airbus A321	200	Oujda
V126	26	105	Fes	Marrakech	06:45:00	07:50:00	26	Asma	Fes	9100.00	105	Boeing 777	350	Meknes
V127	17	104	Oujda	Agadir	08:15:00	09:45:00	17	Walid	Fes	15900.00	104	Airbus A321	200	Oujda
V128	28	107	Marrakech	Rabat	10:00:00	11:15:00	28	Imane	Marrakech	10500.00	107	Airbus A320	180	Casablanca
V129	29	106	Meknes	Fes	12:00:00	12:30:00	29	Youssef	Meknes	14900.00	106	ATR 42	48	Guelmin
V130	10	109	Guelmin	Tanger	14:00:00	16:00:00	10	Sara	Marrakech	7400.00	109	ATR 72	72	Agadir
V131	31	110	Casablanca	Marrakech	15:30:00	16:30:00	31	Khalid	Casablanca	3300.00	110	Airbus A350	300	Tanger
V132	32	101	Rabat	Agadir	17:15:00	18:30:00	32	Lamia	Rabat	11800.00	101	Airbus A350	300	Tanger
V133	13	110	Tanger	Meknes	19:00:00	20:45:00	13	Tarik	Casablanca	3500.00	110	Airbus A350	300	Tanger
V134	34	103	Agadir	Casablanca	21:00:00	22:15:00	34	Hajar	Agadir	19700.00	103	Boeing 747	400	Fes
V135	35	114	Fes	Guelmin	23:00:00	00:30:00	35	Ismail	Fes	14000.00	114	Boeing 777	350	Meknes
V136	26	115	Oujda	Rabat	07:00:00	08:15:00	26	Asma	Fes	9100.00	115	ATR 42	48	Guelmin
V137	37	126	Marrakech	Fes	09:15:00	10:45:00	37	Taha	Marrakech	11200.00	126	Boeing 737	150	Marrakech
V138	38	117	Meknes	Casablanca	11:30:00	12:45:00	38	Dounia	Meknes	3700.00	117	Boeing 737	150	Marrakech
V139	39	128	Guelmin	Marrakech	13:45:00	14:45:00	39	Nabil	Guelmin	17100.00	128	Airbus A350	300	Tanger
V140	4	119	Casablanca	Oujda	15:00:00	16:30:00	4	Yassine	Casablanca	12000.00	119	Airbus A350	300	Tanger
V141	41	120	Rabat	Agadir	17:30:00	18:45:00	41	Zakaria	Rabat	10900.00	120	Falcon 900	12	Rabat
V142	42	121	Tanger	Fes	19:45:00	20:45:00	42	Lina	Tanger	14500.00	121	Boeing 747	400	Fes
V143	3	122	Agadir	Rabat	06:00:00	07:15:00	3	Laila	Guelmin	28500.00	122	Airbus A321	200	Oujda
V144	44	123	Fes	Casablanca	08:30:00	09:45:00	44	Sana	Fes	34000.00	123	Boeing 777	350	Meknes
V145	45	124	Oujda	Tanger	10:45:00	12:15:00	45	Ilyas	Oujda	11700.00	124	ATR 42	48	Guelmin
V146	46	125	Marrakech	Guelmin	13:00:00	14:20:00	46	Nada	Marrakech	5100.00	125	Airbus A320	180	Casablanca
V147	7	126	Meknes	Agadir	15:15:00	16:45:00	7	Zineb	Agadir	5600.00	126	Boeing 737	150	Marrakech
V148	48	127	Guelmin	Fes	17:30:00	19:00:00	48	Rim	Guelmin	14800.00	127	ATR 72	72	Agadir
V149	49	128	Casablanca	Rabat	20:00:00	20:45:00	49	Fouad	Casablanca	10600.00	128	Airbus A350	300	Tanger
V150	50	129	Rabat	Oujda	22:00:00	23:30:00	50	Soukaina	Rabat	30900.00	129	Falcon 900	12	Rabat

49 rows in set (0.001 sec)

18. Pilotes avec salaire > moyenne :

```

MariaDB [bd_air_maroc]> SELECT NOMPIL FROM PILOTE WHERE SALAIRE > (SELECT AVG(SALAIRE) FROM PILOTE);

```

NOMPIL
Amine
Laila
Omar
Noura
Salma
Hamza
Walid
Rayan
Adil
Faycal
Youssef
Brahim
Hajar
Rania
Nabil
Aicha
Lina
Adam
Sana
Reda
Rim
Soukaina

22 rows in set (0.001 sec)

19. Avions capacité > avion 101 :

```
MariaDB [bd_air_maroc]> SELECT * FROM AVION WHERE CAPACITE > (SELECT CAPACITE FROM AVION WHERE NUMAV = 101);
```

NUMAV	Nomav	Capacite	Ville
85	Boeing 747	400	Fes
87	Boeing 777	350	Meknes
94	Boeing 747	400	Fes
96	Boeing 777	350	Meknes
103	Boeing 747	400	Fes
105	Boeing 777	350	Meknes
112	Boeing 747	400	Fes
114	Boeing 777	350	Meknes
121	Boeing 747	400	Fes
123	Boeing 777	350	Meknes
130	Boeing 747	400	Fes

```
11 rows in set (0.001 sec)
```

20. Pilotes sans aucun vol :

```
MariaDB [bd_air_maroc]> SELECT NOMPIL FROM PILOTE WHERE NUMPIL NOT IN (SELECT NUMPIL FROM VOL);
```

NOMPIL
Sami
Houda
Noura
Salma
Hamza
Sofia
Faycal
Jalil
Meryem
Brahim
Rania
Aicha
Adam
Reda

```
14 rows in set (0.001 sec)
```

21. Villes de départ des pilotes de Marrakech :

```
MariaDB [bd_air_maroc]> SELECT DISTINCT VILLE_DEP FROM VOL V JOIN PILOTE P ON V.NUMPIL = P.NUMPIL WHERE P.VILLE = 'Marrakech';
```

VILLE_DEP
Marrakech
Meknes
Guelmin

```
3 rows in set (0.000 sec)
```

22. Vue vols avec noms pilote/avion :

```
MariaDB [bd_air_maroc]> CREATE VIEW VUE_VOL_COMPLET AS SELECT V.*, P.NOMPIL, A.NOMAV FROM VOL V JOIN PILOTE P ON V.NUMPIL = P.NUMPIL JOIN AVION A ON V.NUMAV = A.NUMAV;
```

Query OK, 0 rows affected (0.004 sec)

23. Vue pilotes salaire > 15000 :

```
MariaDB [bd_air_maroc]> CREATE VIEW VUE_PILOTES_RICHES AS SELECT * FROM PILOTE WHERE SALAIRE > 15000;
```

Query OK, 0 rows affected (0.005 sec)

```
MariaDB [bd_air_maroc]>
```


24. Afficher la vue :

```
MariaDB [bd_air_maroc]> SELECT * FROM VUE_PILOTES_RICHE;
```

NUMPIL	NOMPIL	Ville	Salaire
2	Amine	Meknes	31200.00
3	Laila	Guelmin	28500.00
6	Omar	Tanger	19300.00
12	Noura	Guelmin	16200.00
14	Salma	Rabat	18800.00
17	Walid	Fes	15900.00
19	Rayan	Marrakech	30150.00
23	Faycal	Rabat	15200.00
33	Brahim	Tanger	15400.00
34	Hajar	Agadir	19700.00
36	Rania	Oujda	18600.00
39	Nabil	Guelmin	17100.00
40	Aicha	Casablanca	16000.00
43	Adam	Agadir	18300.00
44	Sana	Fes	34000.00
47	Reda	Meknes	19400.00
50	Soukaina	Rabat	30900.00

```
17 rows in set (0.001 sec)
```

25. Supprimer une vue :

```
MariaDB [bd_air_maroc]> DROP VIEW VUE_PILOTES_RICHES;  
Query OK, 0 rows affected (0.001 sec)
```

26. Vue vols départ Marrakech :

```
MariaDB [bd_air_maroc]> CREATE VIEW VUE_VOL_KECH AS SELECT * FROM VOL WHERE VILLE_DEP = 'Marrakech';  
Query OK, 0 rows affected (0.004 sec)
```

27. Créer user_air :

```
MariaDB [(none)]> CREATE USER 'user_air'@'localhost' IDENTIFIED BY '123';  
Query OK, 0 rows affected (0.002 sec)
```

28. Droit de consulter sur PILOTE :

```
MariaDB [(none)]> GRANT SELECT ON bd_air_maroc.PILOTE TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.002 sec)
```

29. Droit d'insertion sur VOL :

```
MariaDB [(none)]> GRANT INSERT ON bd_air_maroc.VOL TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.003 sec)
```

30. Retirer le droit de suppression sur AVION mais avant on donne l'accès a AVION :

```
MariaDB [(none)]> GRANT INSERT ON bd_air_maroc.AVION TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.003 sec)
```

```
MariaDB [(none)]> REVOKE DELETE ON bd_air_maroc.AVION FROM 'user_air'@'localhost';  
Query OK, 0 rows affected (0.003 sec)
```

31. Tous les privilèges notre BD:

```
MariaDB [(none)]> GRANT ALL PRIVILEGES ON BD_AIR_MAROC.* TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.002 sec)
```


32. Créer rôle gestion_vol et attribuer des droits sur la table VOL :

```
MariaDB [(none)]> CREATE ROLE gestion_vol; GRANT SELECT, INSERT, UPDATE ON bd_air_maroc.VOL TO gestion_vol;  
Query OK, 0 rows affected (0.002 sec)  
  
Query OK, 0 rows affected (0.002 sec)
```

33. Affecter le rôle à un utilisateur :

```
MariaDB [(none)]> GRANT gestion_vol TO 'user_air'@'localhost';  
Query OK, 0 rows affected (0.002 sec)
```